SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	U. Sanjay	**Reg. No.:**	1920260111
Date:		**Duration:**	60 minutes

Code of Conduct

I, U. Sanjay(1920260111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U. Sanjay

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

ONLINE FOOD WASTE MANAGEMENT SYSTEM

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

1.1 Project Vision

To develop an accessible online platform that connects restaurants, hotels, cafeterias and households with verified NGOs and volunteers to redistribute safe surplus food. The system will streamline food listings, pickup coordination and donation tracking, helping reduce avoidable food waste while supporting people in need.

1.2 Project Objectives

The main objectives of the Online Food Waste Management System are:

- To provide a convenient online platform for donating safe surplus food from restaurants, hotels, canteens and households.
- To reduce avoidable food waste by publishing available food before its safe collection deadline.

- To connect food donors with verified NGOs and community organizations that can distribute donated food.
- To allow donors to enter food type, quantity, preparation time, location and pickup availability.
- To help NGOs browse nearby listings and reserve suitable food without duplicate bookings.
- To coordinate volunteers and track pickup and delivery status in real time.
- To send timely notifications about new listings, reservations, cancellations and collection updates.
- To maintain a secure database of donor accounts, NGO approvals, listings and donation history.
- To provide dashboards and reports showing donation activity and estimated food quantities redistributed.
- To encourage responsible food handling, transparent coordination and community participation in food redistribution.

1.3 Key Stakeholders and Responsibilities

Stakeholder	Role / responsibility
Food donors	Post accurate surplus food details and prepare food for safe pickup.
NGO coordinators	Verify needs, reserve available food and organize distribution.
Volunteers / delivery partners	Collect food and update pickup and delivery status.
Beneficiaries / communities	Receive redistributed food through participating organizations.
System administrator	Verifies organizations, manages users and moderates listings.
Food safety coordinator	Advises on safe handling and collection deadlines.
Development and QA team	Builds, tests and maintains the application.

1.4 Scope and Assumptions

The first release covers donor and NGO registration, surplus food listings, food requests, pickup coordination, status tracking and an administration dashboard. Payment processing and automated delivery routing are outside Sprint 1. Every listing is assumed to include food type, quantity, preparation time, pickup location and safe collection deadline.

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

2.1 Epic: Surplus Food Listing

US-01: As a food donor, I want to post available surplus food with quantity and pickup deadline, so that nearby organizations can collect it before it goes to waste.

US-02: As a food donor, I want to edit or withdraw a food listing, so that availability remains accurate.

2.2 Epic: Food Requests and Collection

US-03: As an NGO coordinator, I want to browse and reserve available food listings, so that I can organize timely collection for beneficiaries.

US-04: As a volunteer, I want to update pickup and delivery status, so that donors and NGOs can track each donation.

2.3 Epic: Tracking and Reporting

US-05: As a donor, I want to view my donation history, so that I can understand how much surplus food I have shared.

US-06: As an administrator, I want to generate weekly donation reports, so that I can evaluate food redistribution activity.

2.4 Epic: Administration

US-07: As a system administrator, I want to verify NGO accounts, so that food requests are made by approved organizations.

US-08: As a system administrator, I want to manage user roles and flag unsuitable listings, so that the platform remains reliable.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

3.1 US-01 – Surplus Food Listing

Given a registered donor is signed in, when they submit a food type, positive quantity, pickup address and future collection deadline, then the system creates a visible listing with a unique ID. Given a required field is missing, when the donor submits the form, then the system displays a validation message and does not publish the listing.

3.2 US-03 – Food Reservation

Given an available listing has not expired, when a verified NGO reserves it, then the listing status changes to “Reserved” and records the NGO and reservation time. Given the listing is already reserved, when another NGO requests it, then the system prevents duplicate reservations.

3.3 US-04 – Pickup Status Tracking

Given a reserved listing is assigned for collection, when an authorized volunteer marks it “Picked Up”, then the status and timestamp are saved. Given the food reaches the NGO, when the volunteer marks it “Delivered”, then the completed donation appears in the history.

3.4 US-07 – NGO Verification

Given an administrator reviews a pending NGO registration, when the administrator approves it, then the account becomes eligible to reserve listings. Given an account is rejected, when that NGO attempts to reserve food, then the system blocks the request with an explanation.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

4.1 Prioritized Product Backlog

Rank	Story	Priority	Reason / dependency
1	US-07 NGO verification	Must	Establishes trusted recipient accounts.
2	US-01 Food listing	Must	Creates the core donation inventory.
3	US-03 Food reservation	Must	Depends on approved NGOs and active listings.
4	US-04 Pickup tracking	Must	Provides traceability after reservation.
5	US-08 User moderation	Should	Supports account and listing safety.
6	US-02 Edit listing	Should	Improves availability accuracy.
7	US-05 Donation history	Could	Useful after completed donations exist.
8	US-06 Weekly reports	Could	Depends on stored donation records.

4.2 Prioritization Method

MoSCoW prioritization is used: Must-have features enable trusted food listings and reservations; Should-have features improve collection coordination; Could-have features support reporting and insights. Dependencies such as account verification before reservations are considered.

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

5.1 Sprint 1 Overview

Sprint duration: 2 weeks (10 working days). Team assumption: four student contributors. Sprint 1 includes US-07, US-01, US-03 and US-04 using sample donors, NGOs and food listings for demonstration.

5.2 Sprint Backlog and Development Tasks

Story	Actionable tasks	Expected result
US-07	Create NGO profile and verification fields; build admin approval screen; test approval and rejection.	Administrator can verify NGO accounts.
US-01	Design food listing model; build donor form; validate required fields and collection deadline; test listing display.	Donor can publish available food.
US-03	Build listing catalogue and reserve action; prevent double reservation; record reservation timestamp; test conflicts.	Verified NGO can reserve available food.
US-04	Build collection status screen; add pickup/delivery updates; store timestamps; test authorized transitions.	Donation can be tracked to delivery.

5.3 Sprint Workflow and Definition of Done

Sprint Workflow:

The Online Food Waste Management System follows an Agile sprint workflow to support reliable food donation, coordinated pickups and regular review of completed features.

- Sprint Planning: Select high-priority stories and assign food donation workflow tasks to the team.
- Requirement Analysis: Identify donor, NGO, volunteer and administrator needs for safe food listing, reservation and collection.
- System Design: Design registration screens, food listing forms, status tracking and database relationships.

- Development: Build food posting, verified NGO reservations, pickup tracking and administrator workflows.
- Testing: Check form validation, reservation conflicts, pickup status updates and access permissions.
- Sprint Review: Demonstrate the donor-to-NGO donation process and collect stakeholder feedback.
- Sprint Retrospective: Discuss bottlenecks, coordination issues and improvements for the next sprint.

Definition of Done (DoD):

A user story is considered completed when the following conditions are satisfied:

- All specified requirements for the selected user story are implemented.
- Acceptance criteria for food donation workflows are satisfied.
- Food listings display correct quantity, pickup location and collection deadline.
- Reservations and pickup status changes work without duplicate bookings.
- The dashboard displays current available, reserved and completed food donations.
- Donation and collection records are stored securely in the database.
- The system is tested and critical errors are resolved.
- The project team reviews and approves completed features.
- Technical and user documentation is updated.
- The feature is ready for integration and a working demonstration.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

6.1 Estimation Technique

Story points are used in Agile software development to estimate the relative effort required to complete user stories. The estimation considers task complexity, development effort, technical challenges, and potential risks.

In the Online Food Waste Management System, story points help the team plan food listing, reservation, collection and administration features according to complexity and available sprint capacity.

6.2 Story Point Estimates

Story	Points	Justification
US-07	3	Approval form, role validation and database update.
US-01	5	Listing validation, storage, expiry rules and catalogue display.

US-03	5	Reservation logic, conflict prevention and state changes.
US-04	3	Pickup status transitions, timestamps and history.
Total	16	Proposed Sprint 1 workload; revise after team estimation.

6.3 Estimation Risks

User Story	Development Task	Story Points
US01	User login and authentication	3
US02	Surplus food listing	5
US03	Food search and filtering	5
US04	Food reservation management	8
US05	Pickup notifications	5
US06	Donation monitoring dashboard	5
US07	Donation data storage	3
US08	Donation activity reports	5
Total		39

6.4 Justification of Story Point Estimation

- User Authentication (3 Points): Requires donor and NGO sign-in, validation and role checks.
- Food Listing (5 Points): Requires posting forms, food details, expiry checks and listing updates.
- Food Search (5 Points): Requires filtering available listings by location and collection deadline.
- Reservation Management (8 Points): Requires conflict prevention, reservation status and concurrent request handling.
- Pickup Notifications (5 Points): Requires event triggers and timely status messages.
- Donation Dashboard (5 Points): Requires displaying active listings, reservations and donation status.
- Donation Data Storage (3 Points): Requires database tables, relationships and record retrieval.
- Donation Reports (5 Points): Requires aggregating donation history and producing summary reports.

7. Present the Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

7.1 Sprint Goal

The first sprint aims to deliver the core Online Food Waste Management System workflow: NGO verification, surplus food listing, food reservation and pickup status tracking.

The sprint establishes a usable foundation for connecting donors with verified organizations and documenting each donation from listing to collection.

7.2 Expected Deliverables

Deliverables include: a donor food-listing form; verified NGO account approval; available-food catalogue; reservation functionality; pickup and delivery status updates; stored timestamps and donation records; acceptance test results; and a working demonstration with sample data.

7.3 Sprint Review and Success Measures

The sprint is successful when all four selected user stories pass their acceptance tests, duplicate reservations are prevented, and the team can trace a food donation from listing to delivery without unresolved critical defects.

7.4 Proposed Next Sprint

Sprint 2 can add donor history, notifications, volunteer assignment, weekly reporting and enhanced moderation based on stakeholder feedback and team capacity.

Conclusion

The Agile plan turns the Online Food Waste Management System into prioritized, testable development work. The first sprint focuses on verified organizations, accurate food listings, conflict-free reservations and transparent collection tracking. Future sprints can improve coordination and reporting, helping redirect usable surplus food instead of wasting it.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25